



EXP2-4 Blamestorming

Semester B, Weeks 12-14

We all know what **Brainstroming** is - thinking up ideas

Blamestorming is a made-up word from the 'Dilbert' cartoon series by Scott Adams.

Blamestorming is defined as:

“Sitting around in a group discussing why a deadline was missed or a project failed and who was responsible.”

EXP2-4 Blamestroming



The **aim** of EXP2-4 is to consider how to plan a set of experiments to investigate a complex problem in *materials processing*.

The **objectives** for Part 1 of this exercise are:

- Plan a set of experiments to investigate a processing problem
- Work together in your groups
- Write a short report

EXP2-4 What we will do



Part 1: (Week 12-13)

We are going to investigate a materials processing problem associated with the manufacture of earthquake bearings

You are going to work in groups to plan a series of experiments to try and find out what could have caused the process to change

You will present your experimental plan in a Report 1

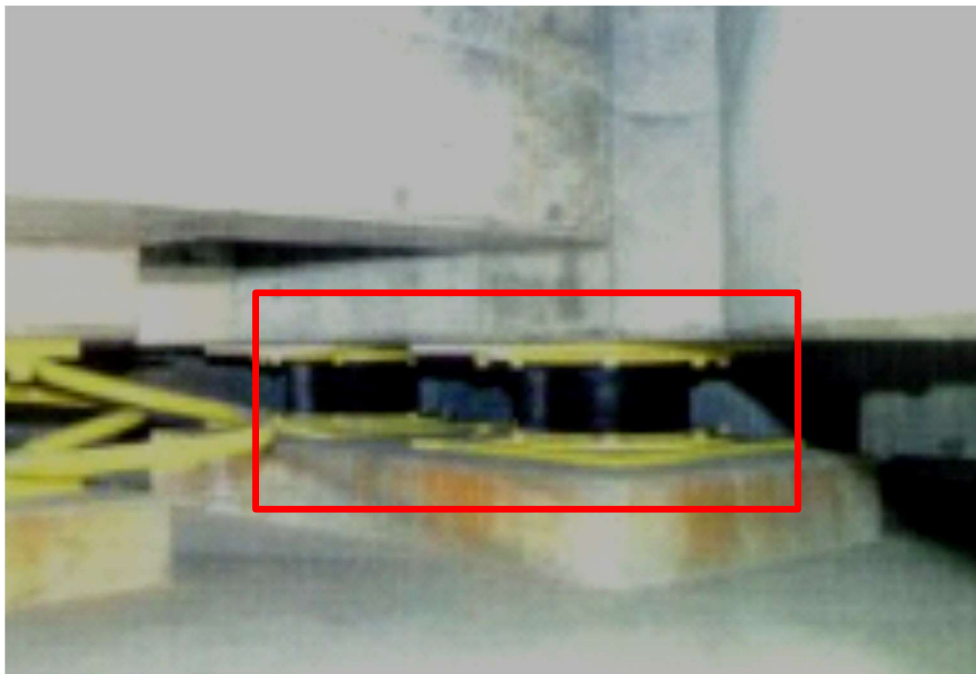
Part 2: (Week 14)

I will give you some experimental data and from some calculations you will be able to tell me what actually changed in Report 2

Introduction to earthquake bearings



In parts of the world where earthquakes are a risk, such as Los Angeles or Tokyo, all buildings over 3 storeys high are stood on earthquake bearings.



- Tower blocks
- Hospitals
- Data centres
- Government buildings

An earthquake is a transverse pressure wave (seismic wave) travelling in the Earth's surface due to movement of fault lines.

Introduction to earthquake bearings



The Damage Caused by Earthquakes



- Collapsed road

- Wenchuan, China 2008



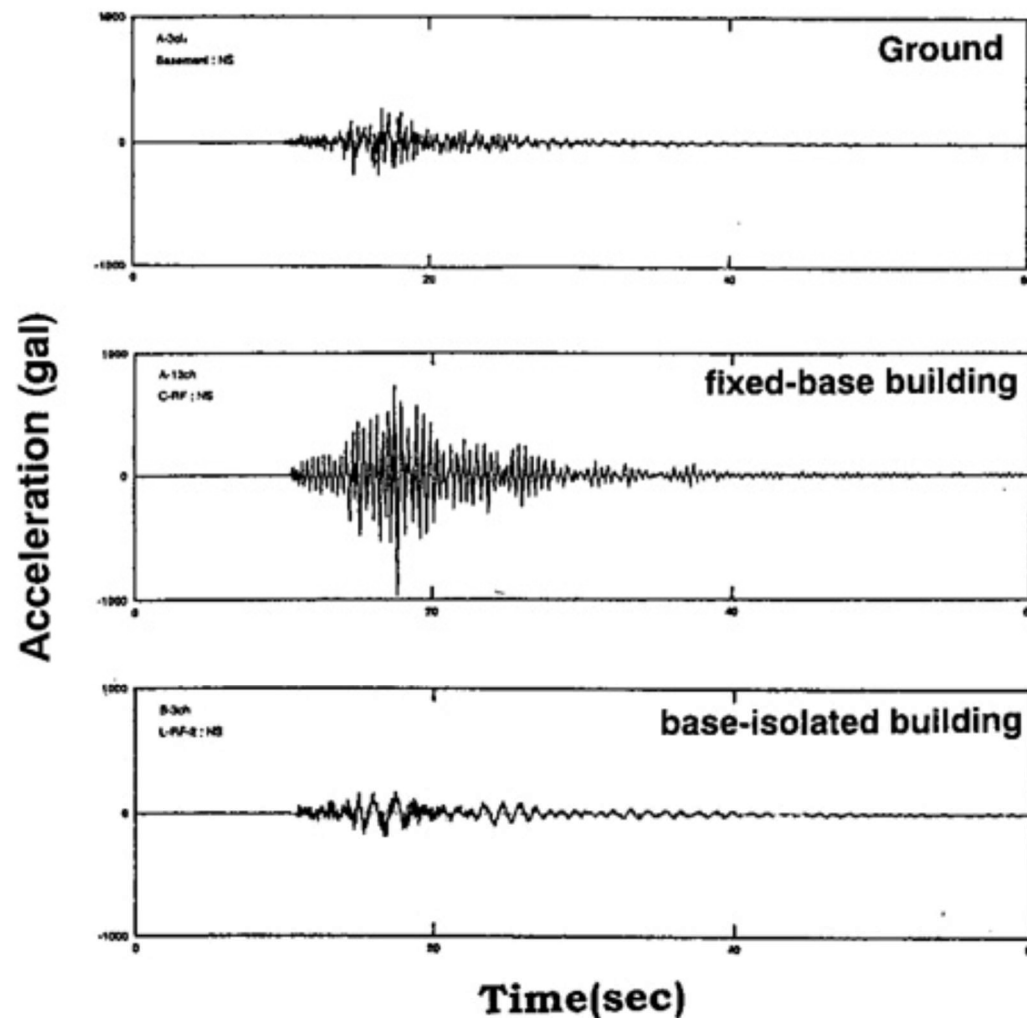
- Domestic Housing

Introduction to earthquake bearings



Why do Structures Collapse?

- Buildings are stiff structures that can resonate if base excited.



Ground Motion

– peak acceleration 0.28g

Rigidly fixed foundations

– peak acceleration 0.99g

Base Isolated building

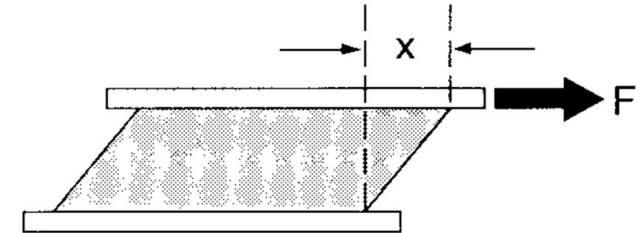
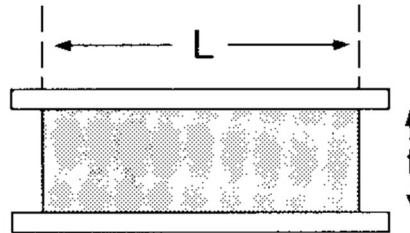
– peak accelerations 0.15g

Introduction to earthquake bearings



Bonded Shear Mountings

Shear modulus, $G = \frac{\tau}{\gamma}$



Shear stress, $\tau = \frac{F}{A}$

Shear strain, $\gamma = \frac{x}{t}$

Shear stiffness = $\frac{F}{x} = \frac{GA}{t}$

A = cross-sectional area

γ = shear strain

τ = shear stress

G = shear modulus

x = shear deflection

F = shearing force

t = thickness of rubber

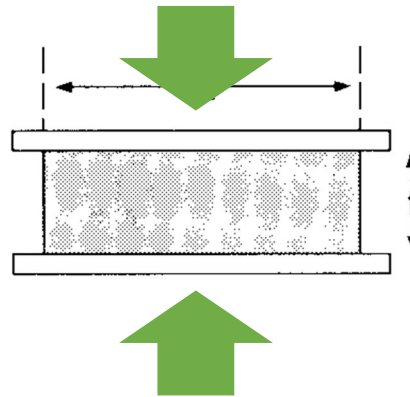
Introduction to earthquake bearings



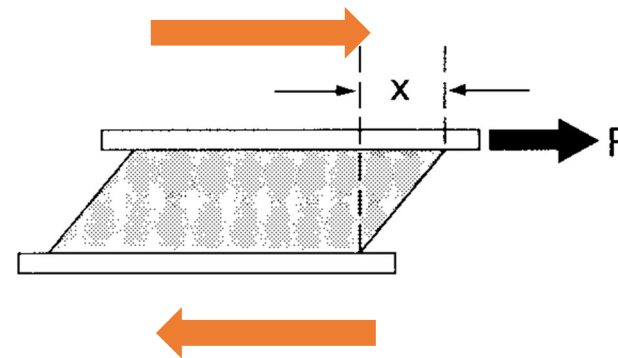
Bonded Shear Mountings

Shear modulus,

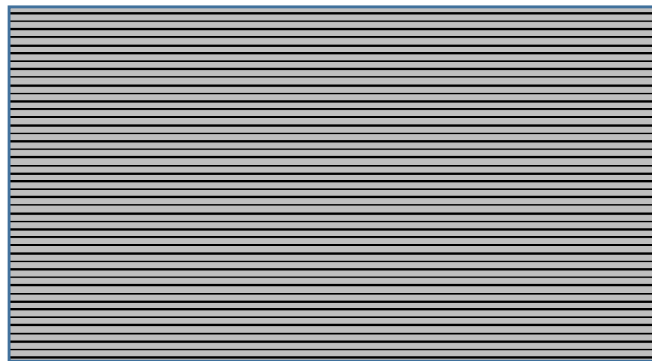
$$G = \tau / \gamma$$



High vertical stiffness



Easy horizontal shear

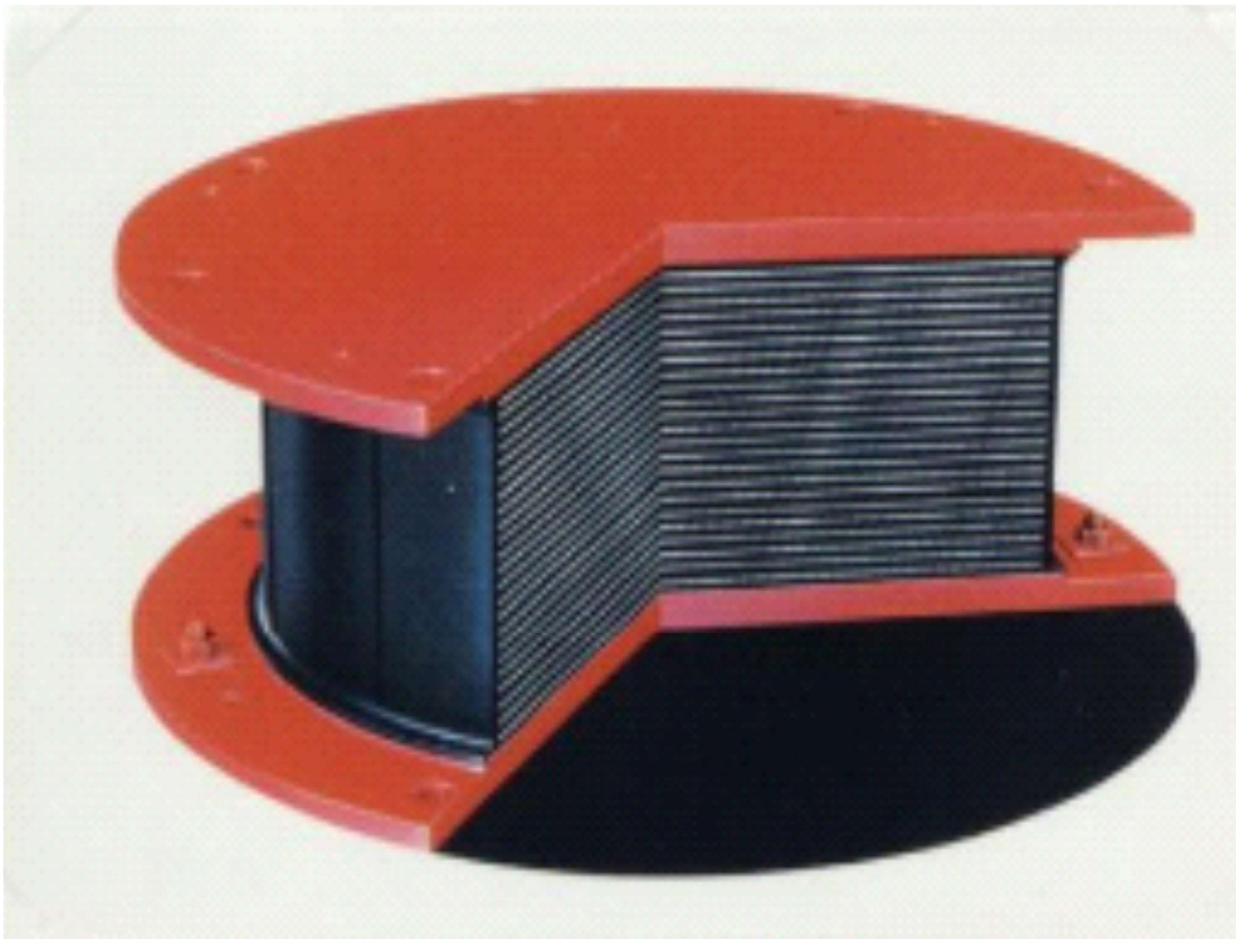


Sheets of steel,
laminated with sheets of
rubber in between

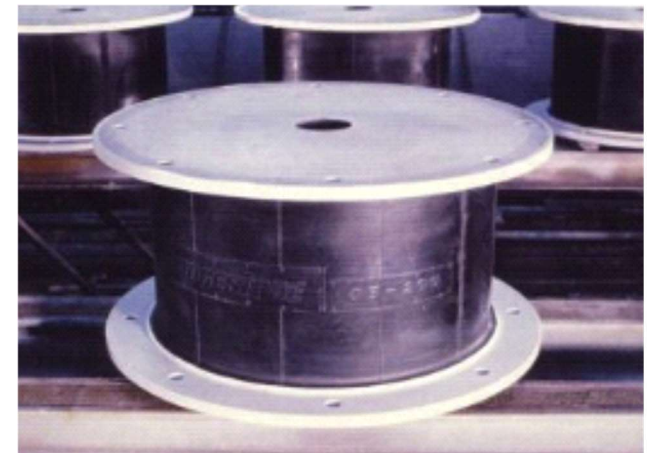
Introduction to earthquake bearings



Base Isolation Mounting Systems



Cross section of a 100 tonne mount



Real parts after manufacture

Introduction to earthquake bearings



Compounding of rubber

- Natural rubbers compounded by mechanical shearing in rolling mill or 'compounding mill'
- Rubber cross-linked by sulphur compounds
- Process catalysed by initiator / accelerator
- Cross-linked by heating (desired elastomeric properties)
- Moulded to shape (desired shear stiffness)

EXP2-2 Blamestorming



We are going to imagine that you work for the *Quakeproof Bearings Company* that manufactures earthquake bearings for large construction projects all over the world

In Report 1 you **do not need to explain**

how an earthquake bearing works

or how to compound, mould and cross-link the rubber

Everything you need to know is in the Blamestorming conversation

Read it carefully!

EXP2-4 Schedule – M Groups



Date:	Time:	Place:	Activity:
Wednesday 7 th	10:30-12:10	A310	Feedback to EXP2-3 Introduction to Part 1 of EXP2-4
Friday 9 th	08:30-10:10		Formulate experimental plan
Monday 12 th	10:30-12:10	A310	Complete experimental plan
Wednesday 14 th	10:30-12:10	QM+	Submit Experimental plan
Friday 16 th	08:30-10:10	A310	Introduction to Part 2 of EXP2-4
Monday 19 th	10:30-12:10	A310	Data processing & analysis
Wednesday 21 st	10:30-12:10		Blamestorming
Friday 23 rd	08:30-10:10		Complete and Submit Report

EXP2-4 Schedule – P Groups



Date:	Time:	Place:	Activity:
Wednesday 7 th	08:30-10:10	A310	Feedback to EXP2-3 Introduction to Part 1 of EXP2-4
Friday 9 th	10:30-12:10		Formulate experimental plan
Monday 12 th	08:30-10:10	A310	Complete experimental plan
Wednesday 14 th	08:30-10:10	QM+	Submit Experimental plan
Friday 16 th	10:30-12:10	A310	Introduction to Part 2 of EXP2-4
Monday 19 th	08:30-10:10	A310	Data processing & analysis
Wednesday 21 st	08:30-10:10		Blamestorming
Friday 23 rd	10:30-12:10		Complete and Submit Report

Planning experiments



How do we plan a programme of experiments?

What do we need to consider?

What is it that you want to know?

Define the problem carefully!

What are the research questions that you are trying to find answers to?

Planning experiments



How do we plan a programme of experiments?

What do we need to consider?

Choose an experiment that will answer
the research question that you have posed

- Will the answer be qualitative or quantitative?
- Will the experiment answer the question completely or will it only partially answer the question?
- Will the experiment exclude a possibility?

The result of an experiment is rarely definitive

Planning experiments



How reliable are the results?

- Can it be repeated consistently?
- Do you always get the same answer?
- Will you obtain precise and accurate results
- or will the error be too great to tell the results apart?
- How many times should the experiment be repeated?

Planning experiments



What resources are needed?

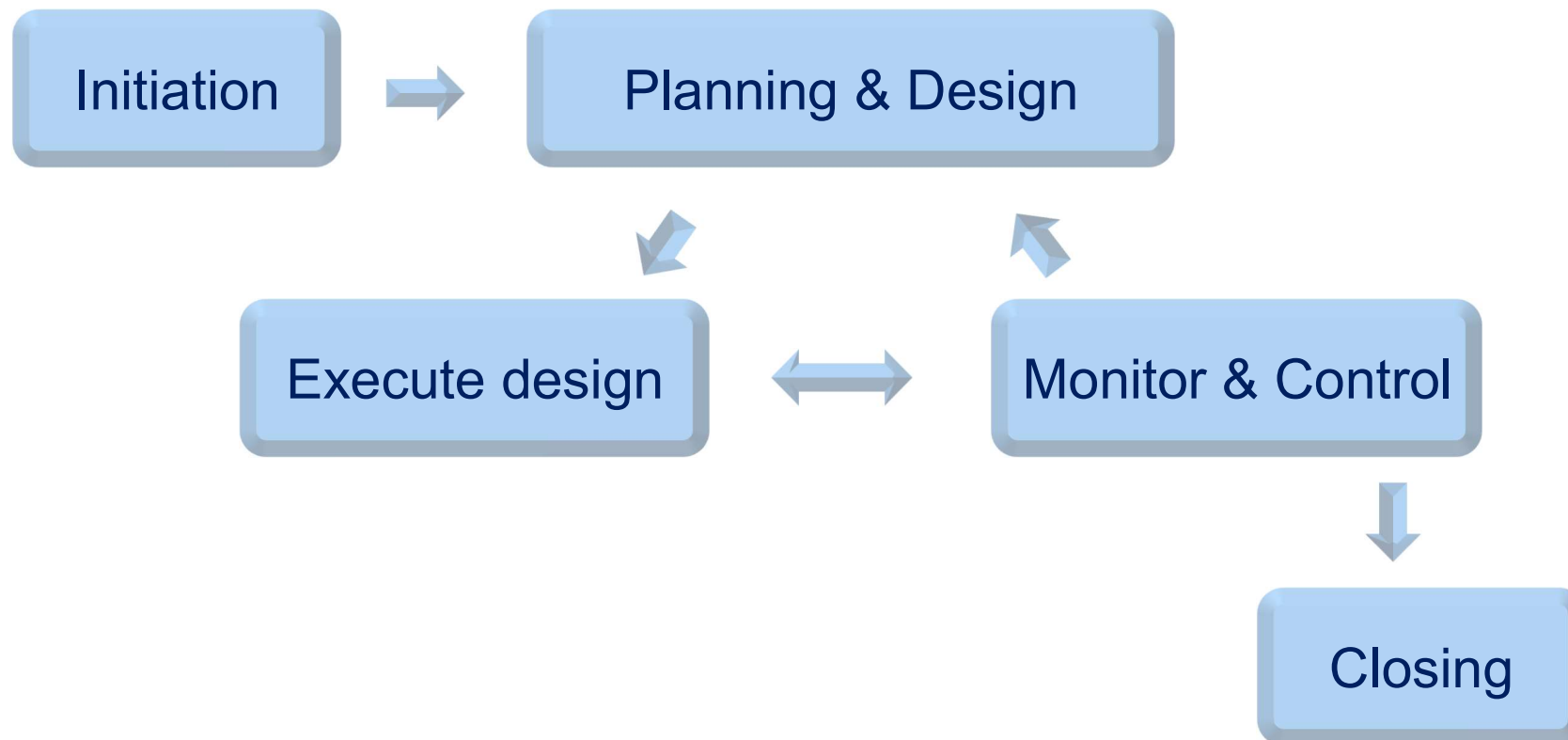
- Is experimental equipment available?
- What people / skills are required to do the experiment will they need training?
- How many samples will be needed?
- How long will it take to get the results?
- What are the risks?

Planning experiments



Project planning tools – flow diagrams

Stages in a project



Planning experiments



Project planning tools – Gantt chart (Henry Gantt 1861-1919)

Timelines of a project

Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Task
M1						Planning
	D1					Specimen manufacture
		M2				Preliminary experiments
				M3		Main experiments
					D2	Report writing

Milestone 1: Project plan completed

Milestone 2: Experimental method developed

Milestone 3: Experimental results obtained

Deliverable 1: Set of specimens made

Deliverable 2: Report completed

Example



What prevents a slice of apple from going brown most effectively?

Define the problem carefully!

Example



What prevents a slice of apple from going brown most effectively?

Define the problem carefully!

- **not** what makes the apple go brown
- what kinds of things could prevent it from going brown?

Example



What prevents a slice of apple from going brown most effectively?

Brainstorming:

What do you already know?

When does an apple go brown?

When does it not go brown?

What are the possible factors?

Example



What prevents a slice of apple from going brown most effectively?

Brainstorming:

Possible factors

Air?

Oxygen?

Oxygen + something else?

Daylight?

Acid?

Alkali?

Heat?

Cold?

Example



What prevents a slice of apple from going brown most effectively?

Brainstorming:

Treatment conditions?

What factors can you control?

What factors can you vary?

What can you combine?

What is possible to do?

Priorities – you can only do a limited number of experiments!

Example



What prevents a slice of apple from going brown most effectively?

Brainstorming:

Measurement?

What are you trying to measure?

Can it be quantified?

How certain is the answer?

How will you compare the slices?

How are you going to record the data?

What is brown? Colour intensity or coverage?

Example



What prevents a slice of apple from going brown most effectively?

Brainstorming:

Experiment plan?

How many samples?

Testing condition?

When and what are you going to measure?

Will the result demonstrate what you are trying to investigate originally?

Example



What prevents a slice of apple from going brown most effectively?

Experimental plan

Control
in Air

Water
T room

Oil

Acid

Alkali

Dark

Water
Fridge

Water
Boiling

Repeat – cut 3 similar apples into 8 pieces

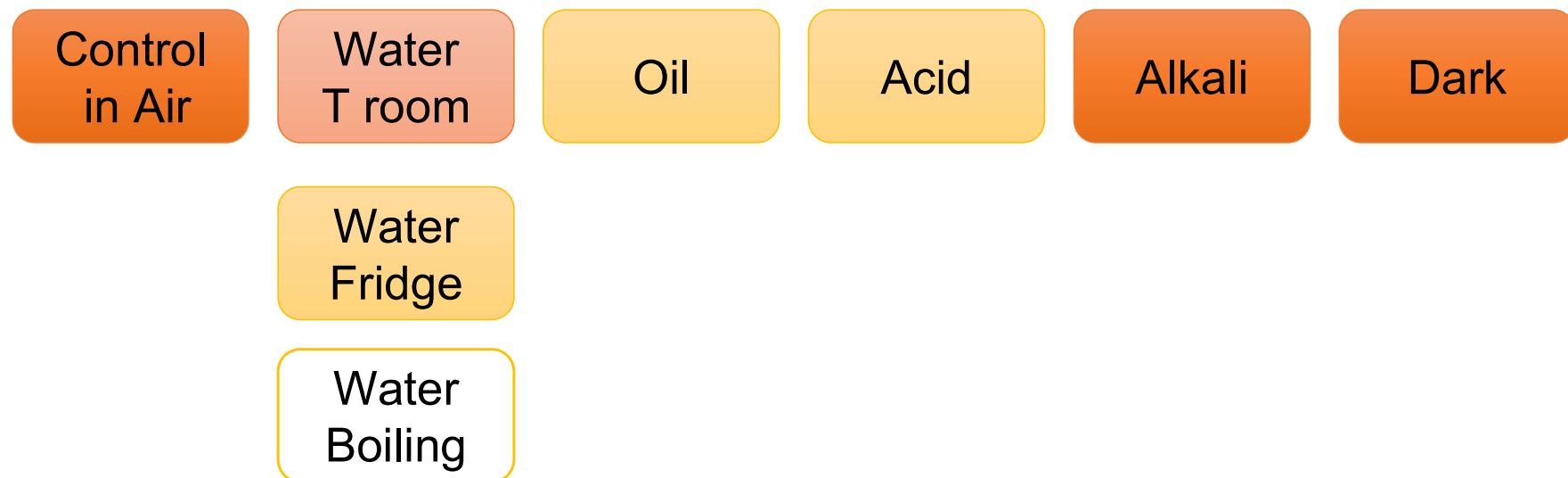
Measure – brownness? Coverage? After what time interval?

Example



What prevents a slice of apple from going brown most effectively?

Result after 1 hour



What did the experiment tell you? Did it answer the question?
Yes, which prevented browning – not what causes the browning